RUDRA PRATAP SINGH RATHORE

Software Engineer

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SUMMARY

B.Tech CSE student (JECRC University) who builds **full-stack applications** and **data-driven ML systems**, from model training to deployment, and aims to ship **production-ready** software. Honest, consistent, and adaptable — quickly picks up new tools, follows through on commitments, and communicates clearly in fast-paced teams.

TECHNICAL SKILLS

Languages: Python, C++, Java, JavaScript, HTML, CSS, SQL
Frameworks: React.js, Node.js, Express.js, REST APIs, MongoDB, Socket.io, Redux Toolkit
Dev Tools: Docker, Git, GitHub, VS Code, Vercel, Linux
ML & AI: scikit-learn, XGBoost, LightGBM, CatBoost, BiLSTM, PPO (Reinforcement Learning), NumPy, Pandas, SciPy
Concepts: Data Structures & Algorithms (DSA), Object-Oriented Programming (OOPs)
Domains: Machine Learning, Data Analysis, Full-Stack Web Development

EDUCATION

B.Tech, Computer Science & Engineering · JECRC University — 2023 – 2027
Senior Secondary (CBSE) · Prince Academy — 2022 – 2023
Secondary (RBSE) · Noble International School — 2020 – 2021

PROJECTS

NEST Prediction System
Python · XGBoost · LightGBM · CatBoost · scikit-learn · JavaScript — Git · Live — 2025

- Engineered a **5-model stacking ensemble** (RandomForest, ExtraTrees, XGBoost, LightGBM, CatBoost) predicting high-frequency NEST exam topics across **5 subjects** and **80 questions** per paper.
- Extracted **55+ temporal, statistical, and era-weighted features** using NumPy, Pandas, and SciPy, encoding recency decay and historical syllabus-shift patterns.
- Deployed a live prediction interface on **Vercel**, integrating a Python ML backend with a JavaScript and HTML/CSS frontend.

Crypto Trade Bot
Python · BiLSTM · PPO (Reinforcement Learning) — Git — 2025

- Developed a **deep RL trading agent** pairing a **BiLSTM** encoder for price-sequence modelling with a **PPO** policy network for trade decision optimization.
- Structured on an active feature branch (bilstm-ppo-improvements), applying iterative architectural refinements to the BiLSTM-PPO pipeline.

Freiwilliger – Volunteer Event Platform
React.js · Node.js · Express.js · MongoDB · Socket.io · Redux Toolkit · JWT — Git — 2026

- Built a **full-stack MERN platform** connecting volunteers with community opportunities across India, featuring location-based event discovery, role-based access, and volunteer profile management.
- Engineered a **Node.js/Express.js REST API** with **JWT authentication**, Firebase phone OTP verification, Cloudinary file storage, and real-time group messaging via **Socket.io** WebSockets.
- Implemented automated reputation scoring, community reviews and ratings, and **Redux Toolkit** state management on a React 18/Vite/Tailwind CSS frontend deployed on Vercel.

INTERNSHIP

Software Engineering Intern · AllsWeb — Certificate — Mar – Jun 2025 (4 months)

- Contributed to software engineering and AI/ML projects at AllsWeb, building scalable web applications, REST APIs, and data-driven solutions.
- Worked on full-stack development, machine learning models, data processing, and deployment in collaborative team environments.

ACHIEVEMENTS AND CERTIFICATES

Smart India Hackathon (SIH) 2024 Round 2 Qualifier, ranked 7th among 200+ college teams; ML Pipeline Developer, Project EDNA — 2024
YUVAi Global Youth Challenge Submitted AI innovation to MeitY & IndiaAI Mission; attended AI bootcamp sessions — 2025
Idea Spark 2.0 Presented Autozentic (AI automation agency) as team Programmer at JECRC — Live — 2025
OpenAI Academy × NxtWave Buildathon Built CO2Watch India, a satellite CO2 monitor for 10 power plants, as Backend Developer — 2026
Google AI Certification (Coursera) Completed a Google-authored AI fundamentals course. — Certificate — 2026